

AI Notes Generator - Updated Project Requirements

Objective

Develop a full stack AI-powered notes generation system where users can generate notes using Artificial Intelligence with custom word limits. The generated notes and chat history should be permanently stored and accessible anytime.

Existing Core Features

- User registration and login using JWT authentication
- AI-powered notes generation using Gemini API
- Permanent storage of chats and generated notes in SQLite
- Dynamic word limit generation based on user requirement
- Notes history management
- Search functionality for previous notes
- Download notes as PDF or TXT
- Responsive React.js frontend with Vite

New Advanced Features Added

1. AI Quiz Generation

- Automatically generates MCQs and quiz questions from generated notes
- Helps students with revision and self-assessment

2. PDF Summarization

- Users can upload PDF documents
- AI summarizes PDF content into short understandable notes

3. Multi-Language Notes

- Generate notes in multiple languages
- Supports English, Telugu, and Hindi

4. Improved Voice-to-Notes Feature

- Converts user voice into text using speech recognition
- Sends converted text to Gemini API for note generation
- Supports voice commands
- Includes automatic language detection
- Includes smart noise reduction
- Includes text-to-speech playback for generated notes

Functional Requirements

- User registration and login
- JWT authentication
- AI notes generation
- Permanent notes storage
- Dynamic word limit support
- Notes history management

- Search notes functionality
- Download notes as PDF/TXT
- AI quiz generation
- PDF upload and summarization
- Multi-language notes generation
- Voice-to-notes conversion
- Text-to-speech playback
- Responsive user interface

Technologies Used

Frontend

- React.js
- Vite
- Tailwind CSS

Backend

- FastAPI

Database

- SQLite

Authentication

- JWT

AI Integration

- Gemini API

Voice Processing

- Speech Recognition API
- Text-to-Speech API

Updated Project Workflow

Step	Process
1	User logs into the system
2	User enters topic, word limit, or uploads PDF
3	User can also use voice input
4	Frontend sends request to FastAPI backend
5	Backend communicates with Gemini API
6	AI generates notes, quizzes, and summaries
7	Generated content stored permanently in SQLite
8	Frontend displays notes and quiz output
9	User can search, download, or replay notes

Expected Output

The final system displays AI-generated notes, quizzes, summarized PDF content, voice-generated notes, notes history, search results, and download options inside a professional dashboard interface.

Unique Features

- Permanent chat and notes storage
- Dynamic word limit generation
- AI-generated quizzes
- PDF summarization
- Multi-language support
- Improved voice-to-notes functionality
- Text-to-speech playback